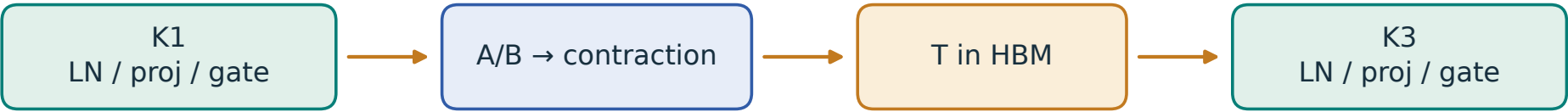


GPU-aware fusion for MiniWorld

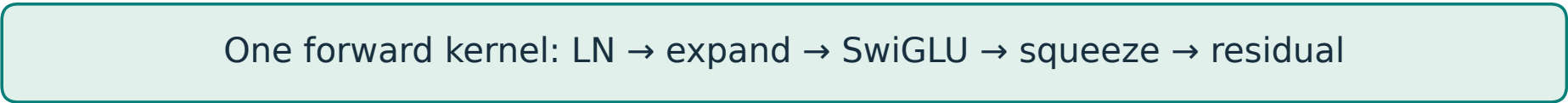
Keep useful intermediates local. Pipeline transfers with compute. Measure forward and backward together.

A / FUSION BOUNDARIES

TriMul · separate global contraction

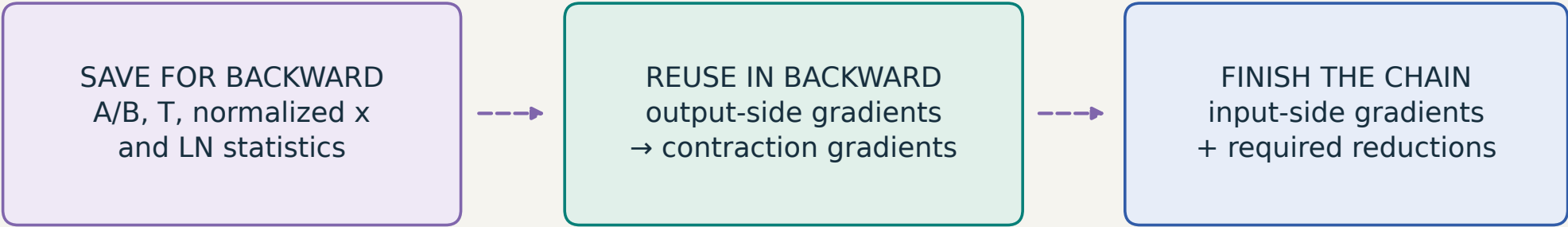
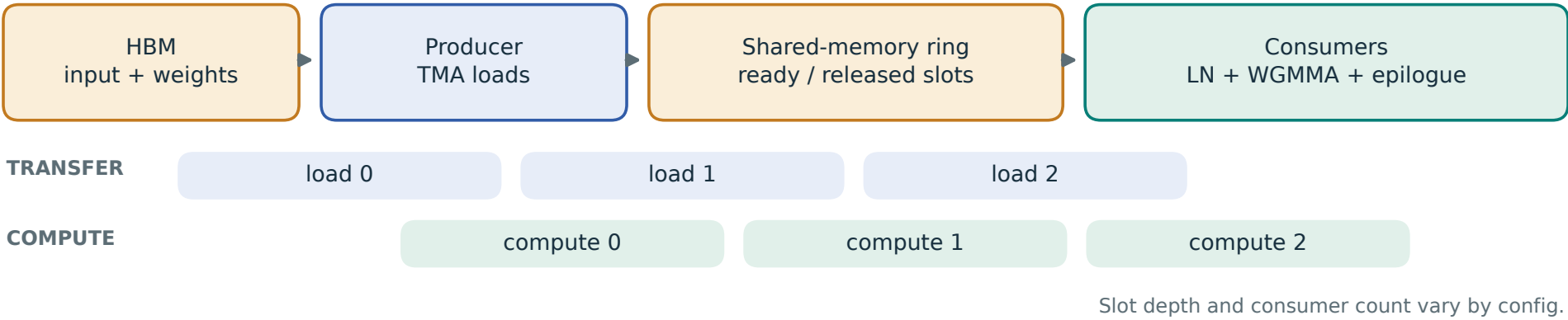


Transition · D128, expansion 4, BF16



B / ALGORITHM AND DATA LIFETIME

INSIDE A K1 / K3 CTA · schematic, not a timing trace



C / MEASURED OPERATOR SPEEDUP

Historical F+B, L384 · family-specific baselines

TriMul **1.61×**

1.634 → 1.018 ms | D128 · T

Transition **1.91×**

1.074 → 0.562 ms | D128, n=4 · X

OPM **2.89×**

5.858 → 2.028 ms | S1024, save O · O

PWA **2.67×**

4.161 → 1.561 ms | S1024 · P

RMSNorm **1.04×**

0.075 → 0.072 ms | D128 · N

LayerNorm **1.12×**

0.276 → 0.246 ms | D384 · N

LNLinear **2.07×**

0.350 → 0.169 ms | D128 → 16; portable path · N

Ratios are not a whole-model speedup.
Portable Norm results do not measure CUDA fusion.

Implementation snapshot: afd54410. Historical checkpoints and baseline identities are documented in the companion figures and evidence bundle.

Fuse operations around data reuse

Mathematical stages at left; audited execution boundaries at right. Supported paths are shape / dtype / GPU specific.

OPERATOR	MATHEMATICAL STAGES · NOT A BASELINE LAUNCH COUNT	IMPLEMENTED GROUPING
TriMul Inference [C1, C3]	LN → input projections / sigmoid gates / mask → contraction → LN / projection / gate / residual	K1 LN + proj + gate + mask → cuBLAS contraction(s) → K3 LN + proj + gate + add A/B and contraction output still cross HBM; bidirectional inference uses two BMM calls.
Transition BF16 D128; n=4 [C4]	LN → expand A/B → SwiGLU → squeeze → residual	One fused forward kernel LN + expand + SwiGLU + squeeze + residual Training saves normalized input / statistics. Backward also has a separate partial reduction.
OPM MSA training [C5]	LN → two projections + mask → outer product → permute / mask-count normalization → output projection	Triton prologue LN + projections + mask → cuBLAS grouped product → CUDA epilogue layout + norm + projection Mask-count normalization is computed separately. Save O or recompute it in backward.
PWA MSA training [C6]	Pair: LN / projection / mask / softmax MSA: LN / value → weighted sum / gate / output / residual	Pair: LN + projection + softmax → MSA: LN + value projection → Weighted sum + gate output + residual First two blocks are parallel input branches, not a serial chain. Training also prepares a dropout mask.
LNLinear CuTe SM90 path [C7]	LN statistics → normalized activation → linear projection	One main kernel GEMM + row statistics + affine correction epilogue Avoids materializing the full normalized activation; weight-side preparation is outside this box.
Triangle Attention Dispatch-dependent [C8, C9]	LN / QKV / bias → tiled attention → sigmoid gate → output projection	Projection path → Tiled attention → Gated output path Dashed boxes denote stages, not single launches. Projection and output fusion depend on dispatch.
LayerNorm RMSNorm Standalone primitives	Row statistics / reduction → normalization + optional affine	Standalone normalization row reduction + normalization / affine A primitive optimization is not evidence of cross-operator fusion. Backward may require partial reductions.

SOLID BORDER audited main kernel / library-call group DASHED BORDER execution stage AMBER ARROW retained global-memory boundary

TriMul: contract globally, reuse locally

A channel-wise contraction connects two fused endpoints. Training adds explicit saved tensors and a separate backward pipeline.

A / THE MATHEMATICS

Outgoing

$$T_{ijc} = \sum_k A_{ikc} B_{jkc}$$

Sum over k; one pair matrix for each channel c.

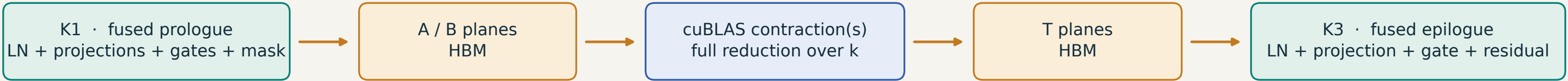
Incoming

$$T_{ijc} = \sum_k A_{kic} B_{kjc}$$

The contracted axis changes; the model semantics stay explicit.

Bidirectional path
separate channel groups
for the two directions

B / THE EXECUTION BOUNDARIES



These HBM intermediates are retained in the packaged inference path. Kernel fusion does not remove the contraction boundary.

INSIDE A K1 / K3 CTA · schematic, not a timing trace

TRANSFER: load 0, load 1, load 2

COMPUTE: compute 0, compute 1, compute 2

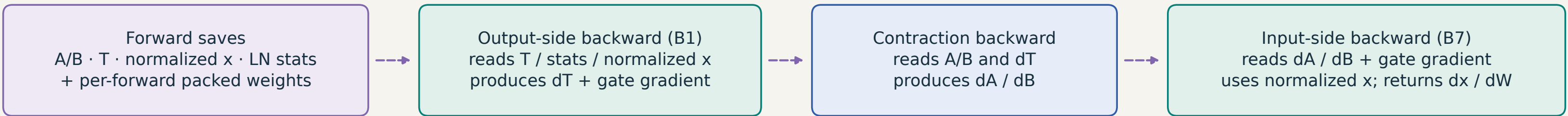
Slot depth and consumer count vary by config.

WHY STOP FUSING HERE?

- Contraction consumes tiles across the k axis.
- CTA-local storage is finite.
- More fusion can raise register / smem pressure.
- A different boundary requires a different algorithm.

The diagram shows current boundaries, not an impossibility proof.

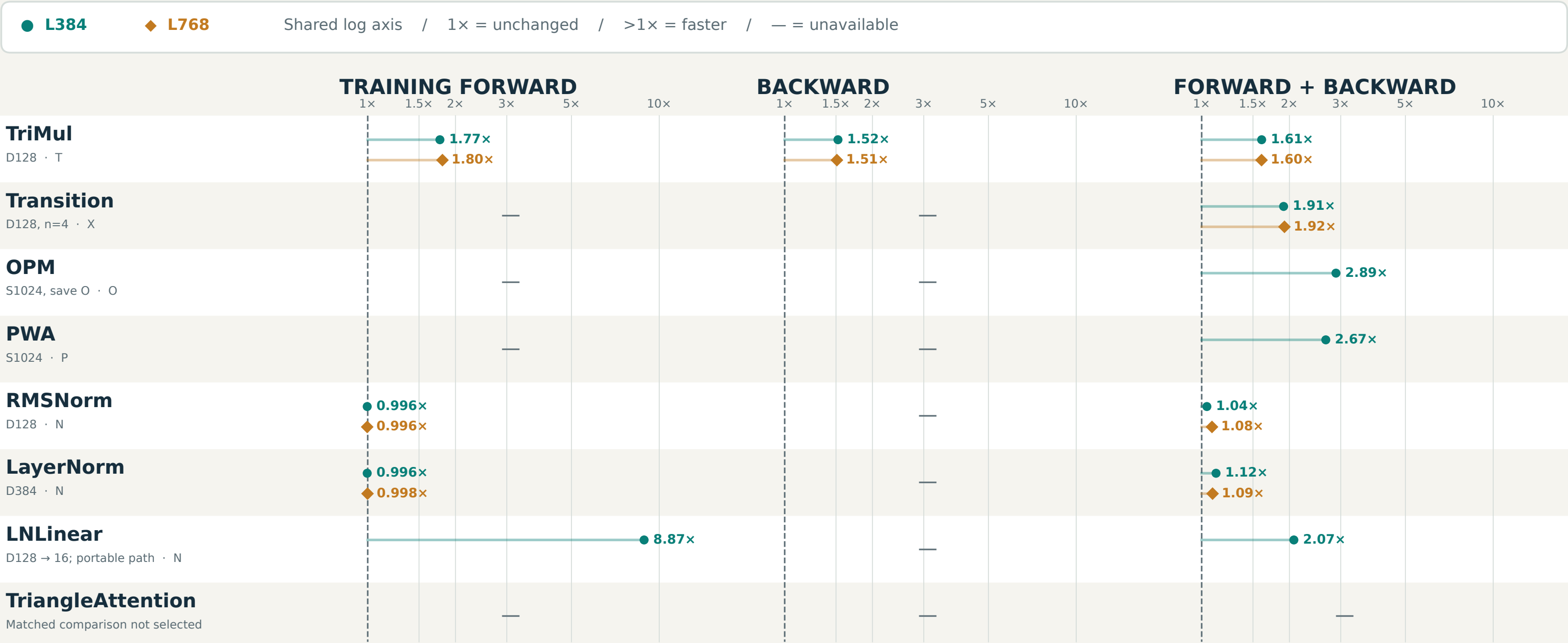
C / TRAINING TENSOR LIFETIMES · D128 PACKAGED PATH



Saved buffers live across forward and backward; gradients and partial reductions may add launches. D256+ research checkpoints are separate.

Measure the full operator, not just the fast stage

Historical H100 records • speedup = baseline time / our time • each source keeps its own baseline • no new GPU measurements



T existing Triton X Triton residual O/P own engine MSA paths N pre-change portable Triton

Forward means training forward here. F+B is measured directly; backward is never inferred by subtraction. No aggregate speedup is computed.

TRIMUL WIDTH SCALING

Research checkpoints for D256+ are not promoted to shared dispatch.

Teal ≥1.5x; blue <1.5x.

	L384 BWD	L384 F+B	L768 BWD	L768 F+B
D128	1.52x	1.61x	1.51x	1.60x
D256	1.27x	1.35x	1.70x	1.66x
D384	1.39x	1.39x	1.53x	1.51x
D512	1.32x	1.35x	1.50x	1.51x